Summary in Graph

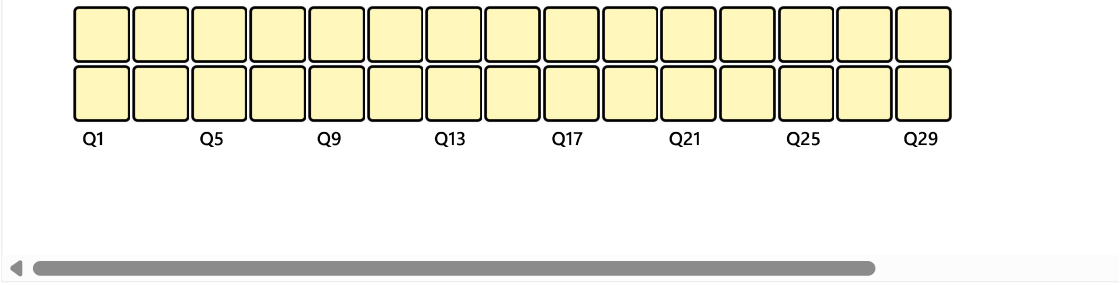
Exam Summary (GO Classes DA Test Series 2026 | Probability and Statistics | Subject Wise Test 2)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**30** (15 + 15)
Total Marks:	**45** (15 + 30)
Exam Duration:	**90 Minutes**
Time Taken:	**0 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (37) out of 368 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X_1, X_2, \ldots, X_{n+1}$ be independent random variables, each distributed as $\text{Bernoulli}(p)$; i.e., for each i,

$$X_i = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

What is the value of

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i X_{i+1}\right]?$$

A. np
B. np^2
C. $(n+1)p^2$
D. n^2p^2

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a random variable with Poisson distribution $p(k;\lambda)$ Let $Y=(X+2)(X+1)$. What is the value of $E(Y)$?

A. $\lambda^2+3\lambda+1$
B. $\lambda^2+3\lambda+2$
C. $\lambda^2+4\lambda+2$
D. $3\lambda^2+3\lambda+2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Which of the following is equal to $\mathrm{Cov}(X+Y, X-Y)$, where X and Y are random variables on a sample space S ?

A. $\mathrm{Var}(X)-\mathrm{Var}(Y)$
B. $\mathrm{Var}\left(X^2\right)-\mathrm{Var}\left(Y^2\right)$
C. $\mathrm{Var}\left(X^2\right)+2\,\mathrm{Cov}(X,Y)+\mathrm{Var}\left(Y^2\right)$
D. $\mathrm{Var}\left(X^2\right)-2\,\mathrm{Cov}(X,Y)+\mathrm{Var}\left(Y^2\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be the number of heads obtained in 40 independent tosses of a fair coin. Then X is a Binomial random variable with

A. $n=40, p=0.$

B. $n=0.5, p=40.$

C. $n=40, p=0.5.$

D. $n=100, p=0.$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

If B_1 and B_2 are mutually exclusive and exhaustive events, then which of the following always holds?

A. $P(B_1 \cup B_2) = 0$.
B. $P(B_1 \cap B_2) = 0$.
C. $P(B_1) = P(B_2)$.
D. $P(B_1) \neq P(B_2)$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let B_1, B_2 be mutually exclusive and exhaustive events, with $P(B_1) = P(B_2) = \frac{1}{2}$. Then for any event A with $P(A \mid B_1) = \frac{1}{4}$ and $P(A \mid B_2) = \frac{3}{4}$, we have

A. $P(A) = \frac{1}{2}$.
B. $P(A) = \frac{1}{4}$.
C. $P(A) = \frac{3}{4}$.
D. $P(A) = \frac{1}{5}$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

If X and Y are discrete random variables, then

A. $P(X = x) = \sum_{y \in \text{Range}(Y)} P(X = x, Y = y)$.
B. $P(X = x) = \sum_{y \in \text{Range}(X)} P(X = x, Y = y)$.
C. $P(X = x) = \sum_{y \in \text{Range}(Y)} P(Y = y)$.
D. $P(X = x) = \sum_{y \in \text{Range}(X)} P(X = x)$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X and Y be jointly continuous random variables with joint density

$$f_{X,Y}(x, y) = \begin{cases} 1 & \text{if } 0 \leq x, y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Then $P(1 < X < 2, 1 < Y < 2)$ is given by

A. 1.

B. 0.5
C. 0.6
D. 0

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 04sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

A rectangle has **length** X and **height** Y, where X and Y are **independent random variables**, each **uniformly distributed** over the interval $[0, 1]$.
Define:

- $A = XY$: the **area** of the rectangle
- $L = 2(X + Y)$: the **perimeter** of the rectangle

Which of the following statements are correct?

A. $\mathbb{E}[A] = \frac{1}{4}$

B. $\mathbb{E}[A] = \frac{1}{2}$

C. $\mathbb{E}[L] = 2$

D. $\mathbb{E}[L] = 1$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let a random variable X be defined by the following process:

- First, flip a fair coin.
- If it lands **heads**, then $X \sim \mathcal{N}\left(\mu_1, \sigma_1^2\right)$.
- If it lands **tails**, then $X \sim \mathcal{N}\left(\mu_2, \sigma_2^2\right)$.

What is the **mean** of the random variable X ?

A. $\mu_1 + \mu_2$

B. $\frac{\mu_1 + \mu_2}{2}$

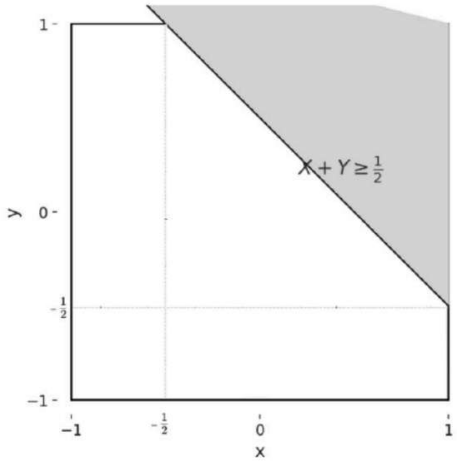
C. $\sqrt{\mu_1^2 + \mu_2^2}/2$

D. $\mu_1 \mu_2$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let (X, Y) be uniformly distributed over the square region $[-1, 1] \times [-1, 1]$. What is the probability that $X + Y \geq \frac{1}{2}$?



A. $\frac{1}{8}$

B. $\frac{1}{4}$

C. $\frac{9}{32}$

D. $\frac{3}{8}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a discrete random variable with probability mass function:

$$f(x) = \binom{3}{x}\left(\frac{1}{4}\right)^{x}\left(\frac{3}{4}\right)^{3-x}, \quad x = 0, 1, 2, 3$$

What is the expected value (mean) of X ?

A. $\frac{3}{4}$

B. $\frac{9}{16}$

C. 1

D. $\frac{3}{2}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X be a discrete random variable with the following probability mass function:

x	-2	3	5
$p(x)$	0.3	0.2	0.5

What is the **variance** of X ?

A. 8.25
B. 7.75
C. 9.25
D. 6.5

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

The length of time (in minutes) for one individual to be served at a cafeteria follows an exponential distribution with a **mean of 4 minutes**.
What is the probability that a person is served in **less than 3 minutes**?

A. $1 - e^{-3/2}$
B. $1 - e^{-3/4}$
C. $1 - e^{-1/4}$
D. $1 - e^{-4/3}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

The lifetime (in years) of a certain type of electrical switch follows an exponential distribution with a **mean of 2 years.**
What is the probability that the switch **fails during the first year**?

A. $1 - e^{-1/4}$
B. $1 - e^{-2}$
C. $1 - e^{-1/2}$
D. $1 - e^{-1}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $X_1, X_2, \ldots, X_6$ be independent random variables such that

$$P(X_i = -1) = P(X_i = 1) = \frac{1}{2}, i = 1, 2, 3, \ldots, 6$$

Then $P\left[\sum_{i=1}^{6} X_i = 4\right]$ is

A. $\frac{3}{32}$
B. $\frac{3}{4}$
C. $\frac{3}{64}$
D. $\frac{3}{16}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A fair 3 -sided die (with faces labeled 1, 2, 3) is rolled three times. Let the random variable X be the **sum of the first two rolls**, and let Y be the **sum of all three rolls.**
What is the value of $\mathbb{P}(X = 5 \mid Y = 7)$?

A. $\frac{2}{9}$

B. $\frac{1}{3}$

C. $\frac{2}{27}$

D. $\frac{2}{6}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A frog starts at position 0 on a one-dimensional number line. Every second, it independently moves one unit to the right with probability p_1, one unit to the left with probability p_2, and stays in place with probability p_3, where $p_1 + p_2 + p_3 = 1$.
Let X be the frog's position after 2 seconds.
Let Y be the movement during the first second, and Z during the second. That is, $X = Y + Z$, and each of $Y, Z \in \{-1, 0, 1\}$.
Using this and the linearity of expectation, what is the value of $\mathbb{E}[X]$ in terms of p_1 and p_2 ?

A. $\mathbb{E}[X] = p_1 - p_2$
B. $\mathbb{E}[X] = 2\,(p_1 + p_2)$
C. $\mathbb{E}[X] = 2\,(p_1 - p_2)$
D. $\mathbb{E}[X] = 2p_3$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let $X_1, X_2, \ldots, X_{n+1}$ be independent random variables, each distributed as $\text{Bernoulli}(p)$.
Define

$$Y = \sum_{i=1}^{n} X_i X_{i+1}$$

Each term $X_i X_{i+1}$ equals 1 if and only if both X_i and X_{i+1} are 1 .

Which of the following statements are **true**?

A. Each term $X_i X_{i+1}$ is a Bernoulli random variable with parameter p^2.

B. The terms $X_i X_{i+1}$ are dependent due to overlapping variables.

C. The expectation of Y is np^2.

D. Each term $X_i X_{i+1}$ is independent, since all X_i are independent.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let E and F be two events with $P(E) > 0, P(F \mid E) = 0.3$ and $P\left(E \cap F^c\right) = 0.2$. Then $P(E)$ equals

A. $\frac{1}{7}$

B. $\frac{2}{7}$

C. $\frac{4}{7}$

D. $\frac{5}{7}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X be a discrete random variable that takes exactly two distinct values, has **mean 10**, and **variance 9** . Assume both values are taken with equal probability.

Which pair of values does the random variable X take?

A. $\{7, 13\}$

B. $\{8, 12\}$

C. $\{6, 14\}$

D. $\{5, 15\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Two coins with probability of heads u and v, respectively, are tossed independently. If P (both coins show up tails) $= P$ (both coins show up heads), then $u + v$ equals

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{3}{4}$

D. 1

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let X_1, X_2, X_3 and X_4 be independent random variables. Then which of the following pairs of random variables are independent?

A. $(X_1 + X_2, X_2 + X_3)$

B. $(X_1, X_1 + X_3)$

C. $(X_1 + X_2, X_3)$

D. $(X_2, X_1 + X_2 + X_3)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose A and B are events with $P(A) = 0.5, P(B) = 0.4$ and $P\left(A \cap B^c\right) = 0.2$. Then $P\left(B^c \mid A \cup B\right)$ is equal to

A. $1/2$

B. $1/3$

C. $1/4$

D. 0

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let E and F be two events with $0 < P(E) < 1, 0 < P(F) < 1$ and $P(E \mid F) > P(E)$. Which of the following statements is (are) TRUE?

A. $P(F \mid E) > P(F)$

B. $P\left(E \mid F^c\right) > P(E)$

C. $P\left(F \mid E^C\right) < P(F)$

D. E and F are independent

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

An insurance company determines that N, the number of claims received in a week, is a random variable with $\mathrm{P}[N = n] = \frac{1}{2^{n+1}}$, where $n \geq 0$. The company also determines that the number of claims received in a given week is independent of the number of claims received in any other week.
Determine the probability that exactly seven claims will be received during a given two-week period.

A. $\frac{1}{256}$

B. $\frac{1}{128}$

C. $\frac{7}{512}$

D. $\frac{1}{64}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A miner is trapped in a mine with **three doors**:

- **Door 1** leads to **safety** in **2 hours**.
- **Door 2** leads back to the mine after **3 hours**.
- **Door 3** leads back to the mine after **5 hours**.

Every time the miner faces the doors, he picks one **uniformly at random** (i.e., with equal probability). What is the expected time (in hours) for the miner to r**each safety**?

A. 6 hours
B. 9 hours
C. 10 hours
D. 12 hours

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

The jobs arrive at a facility randomly over time. The number of jobs arriving in a fixed time interval follows which distribution?

A. Normal
B. Poisson
C. Exponential
D. Uniform

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A coin is tossed thrice. Let X be the event that head occurs in each of the first two tosses. Let Y be the event that a tail occurs on the third toss. Let Z be the event that two tails occur in three tosses. Based on the above information, which one of the following statements is TRUE?

A. X and Y are not independent

B. Y and Z are dependent

C. Y and Z are independent

D. X and Z are independent

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $\{1, 2, 3, 4\}$ represent the outcomes of a random experiment, and $P(\{1\}) = P(\{2\}) = P(\{3\}) = P(\{4\}) = 1/4$. Suppose that $A_1 = \{1, 2\}$, $A_2 = \{2, 3\}$, $A_3 = \{3, 4\}$, and $A_4 = \{1, 2, 3\}$. Then which of the following statements is true?

A. A_1 and A_2 are not independent.

B. A_3 and A_4 are independent.

C. A_1 and A_4 are not independent.

D. A_2 and A_4 are independent.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

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